# Problem Charter: Waris

*An agentic AI system for unclaimed inheritance recovery*

| Project details | |
|---|---|
| Course | Agentic AI (26CD7PEAAI) · B.E. CSE (Data Science) · Semester VII |
| Department | CSE (Data Science), BMS College of Engineering |
| Instructor | Dr. Kalyan Nagaraj |
| Review | Review 1: Problem and Area Defense · 16 October 2026 |
| Team Lead | Srijan C Vachadmath |
| Members | Chirag T · Manohara Salmani · Mohammed Arshaq |

## 1. Niche Area

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|---|---|
| Sector | Personal finance and legal-tech |
| Niche | Unclaimed inheritance recovery for families of deceased persons |
| Scale | Unclaimed financial assets exceeded ₹1.1 lakh crore in FY26 [1] |
| Recovery gap | Banks returned only ₹10,297 crore between April 2022 and November 2025 [2] |
| Urgency | Supreme Court sought responses from the government and SEBI on informing heirs, May 2026 [2] |
| Why not obvious | Not a chatbot, recommender or document Q&A; no existing tool performs the claim itself |

# 2. Problem Statement

> Families cannot reliably discover and claim the financial assets a deceased relative left behind, because each bank, insurer and share registry keeps separate records, rules and document requirements, and small mismatches in names or details stop claims.

| Root cause | Description |
|---|---|
| Unawareness | Families often do not know the assets exist |
| Fragmentation | Each institution has its own portal, procedure and documents |
| Mismatches | One difference, e.g. "R. Subramanian" vs "Ramaswamy Subramanian", can halt a claim |
| Existing gap | The government's portal (June 2026) supports search only, not claiming |

# 3. Decision Supported

> **For each asset left behind: is the claim ready to file, and if not, exactly what is missing?**

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| Input | Deceased person's details and scanned documents provided by the family |
| Output | Per-asset status (Ready · Missing items · Needs human review), document checklist and draft claim letters |
| Final authority | The family; Waris is advisory and never moves money, files claims or decides heirship |

# 4. Agentic Design and Concept Mapping

### 4.1 Agent loop (how one case runs)

1. **Plan:** the Supervisor splits the case into tasks, one per asset type.

2. **Act:** the Tracer chooses and calls tools: reads scanned papers, searches portals, matches names.
3. **Retrieve:** the Rules Expert fetches the claim procedure that fits the case, grades it, and re-queries or abstains if it does not fit.
4. **Build:** the Builder drafts the document checklist and claim letters.
5. **Verify:** the Verifier checks every item against evidence and sends weak work back to step 2, 3 or 4 (at most 6 rounds).
6. **Confirm:** the agent pauses for the family on uncertain matches and before releasing the final pack.
7. **Stop:** when every asset is marked Ready, Missing items or Needs human review, or the round limit is reached.

### 4.2 Concept mapping, with an example from one case

| Concept | How it is used in Waris | Example |
|---|---|---|
| Tool use (mandatory) | The agent decides which tool to call and acts on the result | Reading a passbook photo returns "R. Subramanian, A/c 4471"; a portal search then finds the matching fixed deposit |
| Agentic RAG (mandatory) | Claim procedures are stored as searchable documents; the agent builds a query from case facts, grades each result for fit, then re-queries or abstains | First result is a nominee rule, but the family are legal heirs: graded unfit, query refined, correct procedure retrieved |
| Planning loop (mandatory) | Plan-and-execute, with think-act-observe steps and a Verifier reflection loop | Verifier finds no proof of the name difference in the checklist; Builder adds it and resubmits |
| Observability (mandatory) | Every decision, tool call, retrieval and score is logged as a trace in Langfuse | The trace shows exactly where the agent re-queried and why |
| Evaluation (mandatory) | Outputs compared with a 60-case golden dataset | Asset recall, checklist accuracy, correct abstentions, and zero wrong-person matches |

| Concept | How it is used in Waris | Example |
|---|---|---|
| Multi-agent (bonus) | Supervisor routes work to four specialist agents | Tracer finds assets while Rules Expert fetches procedures |
| Memory (bonus) | Working memory for the current case; long-term case file across visits | Family returns a week later with an affidavit; work continues from where it stopped |
| Human-in-the-loop and guardrails (bonus) | Approval pauses, input checks, masked personal data, capped loops and model calls | A share record matches the name but not the address, so the agent asks before continuing |
| Durable workflow (bonus) | Case state is checkpointed after every step | After a restart, the case resumes from the last saved step |
| Experimentation (bonus) | A/B test and removal tests, repeated 3 times | Single-pass RAG baseline vs full Waris; then Waris without the Verifier or without re-querying |

## 5. Feasibility Plan

| Item | Plan |
|---|---|
| Data | Seeded synthetic families, portal records and scanned documents with known answers; no real personal data |
| Knowledge sources | Public RBI, IRDAI and IEPF claim procedures, cited |
| Technology (all free) | LangGraph · Llama via Ollama (local) · Chroma · FastAPI with SQLite · Tesseract · RapidFuzz · Langfuse · Streamlit |
| Scope | 3 asset types (bank deposits, life insurance, shares); 3 test portals; 60 test cases |
| Out of scope | Real portals, real submissions, legal heirship decisions |

| Item | Plan |
|---|---|
| Safety | Advisory output, human confirmation, cited rules, abstention under uncertainty, capped loops and model calls |

### Team roles

| Member | Responsibility |
|---|---|
| Srijan C Vachadmath (Lead) | Supervisor and agent loop, coordination, demo |
| Chirag T | Test portals, data generator, name matching, document reading |
| Manohara Salmani | Rule corpus, retrieval and re-query logic |
| Mohammed Arshaq | Evaluation, tracing, experiment, report |

### Milestones

| Review | Date | Deliverable |
|---|---|---|
| R2 | 24 Oct 2026 | Design document and architecture |
| R3 | 31 Oct 2026 | Working prototype with traces |
| R4 | 7 Nov 2026 | Final system, evaluation and experiment |
| Submission | 12 Nov 2026 | Report, slides and code repository |

## 6. Backup Idea

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| Title | Hospital Claim Recovery |
| Decision supported | Which items in a hospital bill will the insurer likely deduct, and what fixes the claim before filing? |
| Evidence | Insurers disallowed or repudiated ₹26,000 crore of health claims in FY24 [3] |
| Safety | Billing only, no medical advice; billing staff approve every output |

# References

1. Lapaas Voice, Unclaimed assets cross ₹1.1 lakh crore in FY26, citing government data to the Lok Sabha.
2. Moneylife, Supreme Court seeks response on informing heirs, 6 May 2026.
3. Cafemutual, Health insurance claim rejections in FY24, citing IRDAI data.